POWER BI · LESSON 1 OF 6

Getting Started with Power BI & Your First HR Report

Understand what Power BI is and how its parts fit together, install Power BI Desktop, tour the three views, follow the ETL and analytics stages, and build a complete first report on HR data: cards, bar and column charts, a donut, slicers, a year column with DAX, groups and small multiples.

Power BI class 1 | Power BI Desktop 2026 (Copilot, Fabric, PBIP) | Beginner | 120 min

By the end of this lesson you can

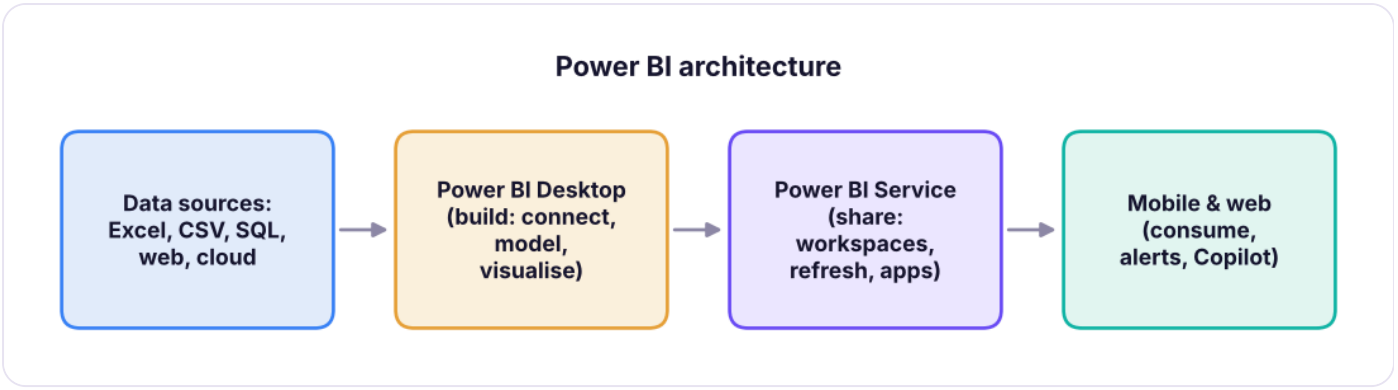
- Describe Power BI Desktop, the Service and Mobile, and how data flows between them
- Explain ETL and the stages of the data-analytics process, and map them to Power BI tools
- Navigate Report, Table and Model views, and the Fields, Visualizations and Filters panes
- Load a CSV, check data types and build cards, bar/column charts, a matrix, a donut and slicers
- Add a calculated column with DAX (YEAR), create data groups, and use small multiples

Datasets used

`hr_employees.csv`

What is Power BI?

Power BI is Microsoft's business-intelligence platform: it connects to data, cleans and models it, and turns it into interactive reports and dashboards that people can share. In 2026 it is part of **Microsoft Fabric**, Microsoft's all-in-one analytics platform, and it is one of the most requested tools in analyst job posts, alongside Excel, SQL and Tableau.



The Power BI components. You build in Desktop, publish to the Service, and consume on the web or mobile.

Component	What it is	Who uses it
Power BI Desktop	Free Windows application to build data models and reports (`.pbix`)	Analysts, developers
Power BI Service (app.powerbi.com)	Cloud site to publish, share, schedule refresh, create dashboards and apps	Analysts and report consumers
Power BI Mobile	iOS/Android apps to view reports and get alerts	Managers on the go
Power BI Report Builder	Paginated, print-ready reports (invoices, statements)	Report developers
On-premises data gateway	Secure bridge so the cloud can refresh data from your own servers	Admins
Power BI Report Server	Self-hosted option for organisations that cannot use the cloud	IT
Copilot	AI assistant that writes DAX, builds report pages and summarises insights (needs a Fabric/Premium capacity)	Everyone

TIP

Report vs dashboard

In Power BI a **report** is one or more pages of interactive visuals built on a data model. A **dashboard** (Service only) is a single page of **pinned tiles** from different reports. Many teams use reports only, or share them as an **app**.

Licences in brief

Licence	Use
Power BI Desktop	Free: build and save locally
Free (Service)	Personal use in "My workspace"; cannot share with others
Pro	Publish and share with other Pro users (about US$14 per user per month)
Premium Per User / Fabric capacity	Larger models, AI features, share with free users

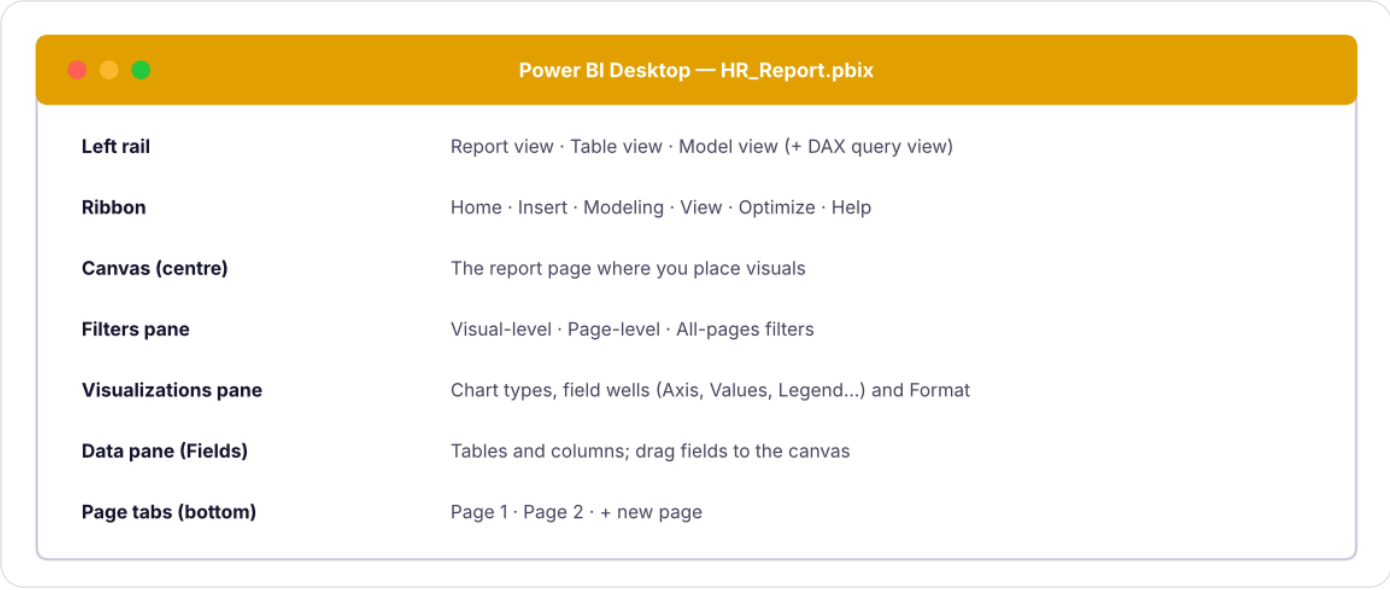
(Prices change; check Microsoft's current pricing page.)

Installing Power BI Desktop

STEP BY STEP **Install**

1. **Easiest:** open the **Microsoft Store**, search "Power BI Desktop" and click Install (it updates itself every month).
2. Or download it from the Microsoft Power BI website → Download → Power BI Desktop.
3. Requirements: Windows 10/11 (64-bit), 4 GB+ RAM (8 GB+ recommended). **There is no Mac version**: Mac users can use the Service in a browser, a Windows virtual machine, or Windows 365.
4. Start the app and sign in with a work or school account (optional for building locally).

The three views

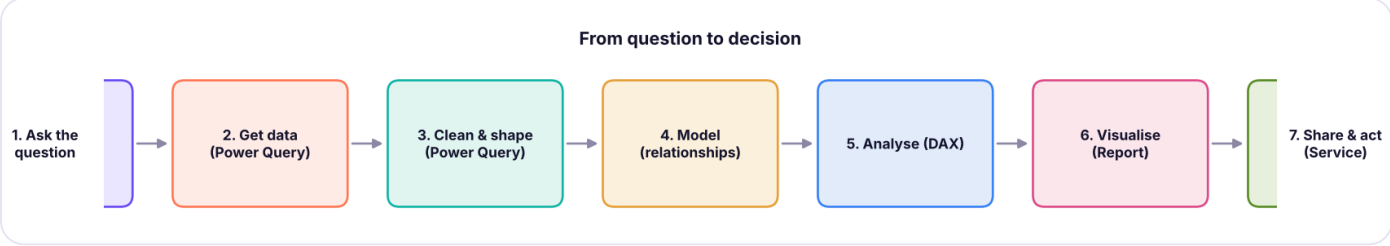


The Power BI Desktop window: left rail with three views, canvas in the middle, panes on the right.

View	Icon	Use it to
Report	bar chart	Design the report pages
Table (Data)	grid	Inspect the data and create calculated columns and measures
Model	diagram	See and edit **relationships** between tables
DAX query view	`{ }`	Run DAX queries and test measures

ETL and the analytics process

ETL = **E**xtract (get data from sources), **T**ransform (clean and reshape it), **L**oad (store it in the model). In Power BI, Extract and Transform are done in **Power Query**, and Load puts the result into the data model. (You met Power Query in Excel Lesson 12: it is the same engine.)



The data analytics stages, mapped to Power BI.

Always start with the **question**: "Why are people leaving?" decides which data and visuals you need. The worst Power BI reports are tours of every chart type with no question.

Your first report: HR attrition

CASE STUDY **The brief**

HR wants a one-page report on the company's 1,500 employees: how many there are, how many left (attrition), where attrition is highest, and whether overtime, job satisfaction or income play a role. Use `hr_employees.csv`.

The data (first rows):

Python

```
import pandas as pd
hr = pd.read_csv("hr_employees.csv", parse_dates=["hire_date"])
print(hr.shape)
hr.head(5)
```

OUTPUT

```
(1500, 15)
```

	emp_id	gender	age	department	job_title	education	hire_date	years_at_c
0	E00001	Male	39	Operations	Operations Manager	Bachelor's	2025-05-26	0
1	E00002	Female	45	IT	Tester	Bachelor's	2020-10-17	5
2	E00003	Male	41	IT	Tester	Bachelor's	2019-05-06	6
3	E00004	Male	21	R&D	Data Scientist	Bachelor's	2025-08-23	0
4	E00005	Male	41	Sales	Sales Executive	Master's	2015-08-24	10

5 rows × 15 columns

Step 1: get the data

STEP BY STEP **Load the CSV**

1. Home → **Get data → Text/CSV** → choose `hr_employees.csv`.
2. The preview shows the columns and detected data types. Click **Transform Data** (not Load) to check types in Power Query.
3. Confirm: `age`, `years_at_company`, `monthly_income`, `job_satisfaction`, `performance_rating` are whole numbers; `hire_date` is **Date**; the rest are text.
4. Home → **Close & Apply**.

In the Data pane you now see the table `hr_employees` with a **Σ** next to numeric fields (they can be summed) and no symbol next to text fields.

Step 2: KPI cards

Drag fields onto the canvas, then choose the **Card** visual:

- **Employees** = Count of `emp_id`
- **Attrition (leavers)**: use a measure (Lesson 3 explains measures): `Leavers = CALCULATE ( COUNTROWS ( hr_employees ), hr_employees[attrition] = "Yes" )`
- **Attrition %** = `DIVIDE ( [Leavers], COUNTROWS ( hr_employees ) )`, formatted as a percentage
- **Average monthly income** = Average of `monthly_income`

Python

```
n = len(hr); leavers = (hr.attrition == "Yes").sum()
print(f"Employees      : {n:,}")
print(f"Leavers        : {leavers:,}")
print(f"Attrition %    : {leavers / n:.1%}")
print(f"Avg income (₹) : {hr.monthly_income.mean():,.0f}")
```

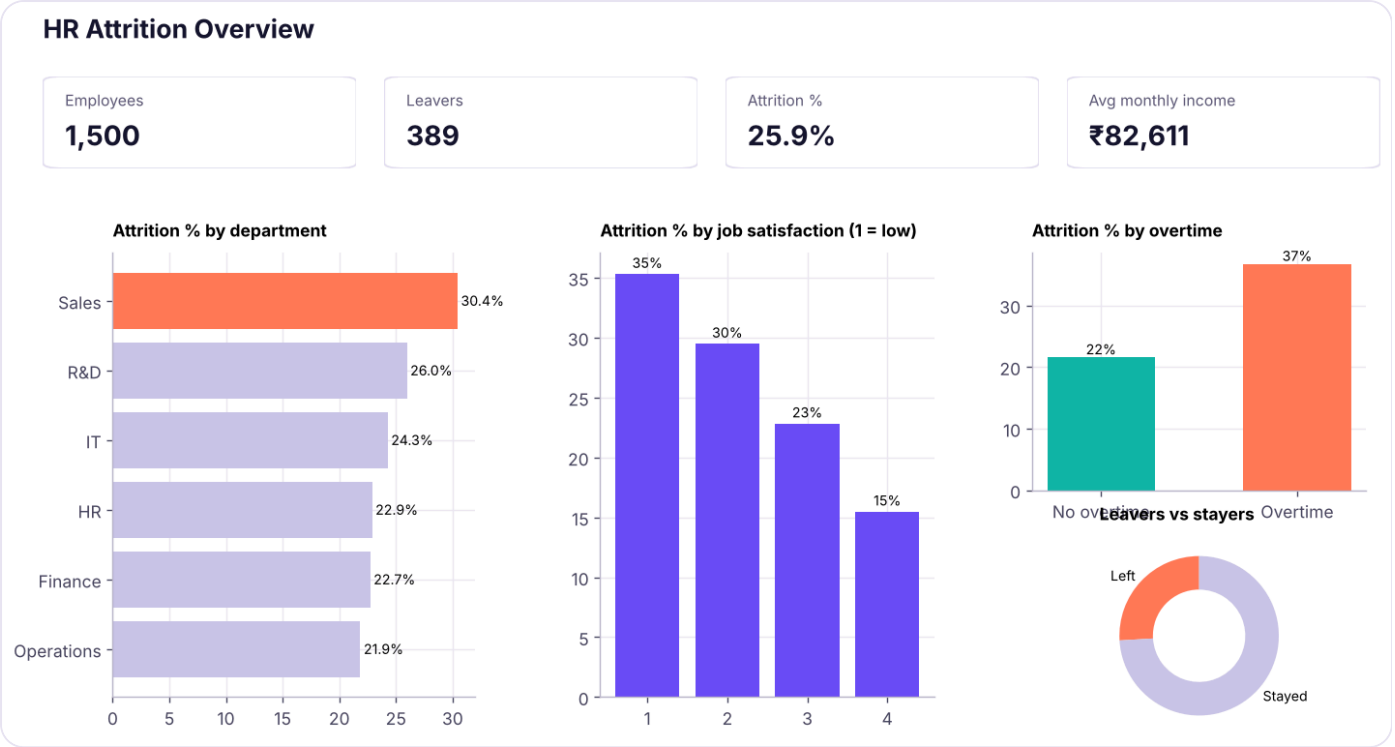
OUTPUT

```
Employees      : 1,500
Leavers        : 389
Attrition %    : 25.9%
Avg income (₹) : 82,611
```

Step 3: bar, column and donut charts

- **Clustered bar chart**: Y-axis `department`, X-axis = `Attrition %` → ranks departments.
- **Clustered column chart**: X-axis `job_satisfaction`, Y-axis `Attrition %`.
- **Donut chart**: Legend `attrition`, Values = count of `emp_id`.

- **Matrix**: Rows `department`, Columns `overtime`, Values `Attrition %` (turns into a heat map with conditional formatting, Lesson 5).
- **Slicers**: `gender`, `education`, `city`.



The finished HR page, drawn from the real data: KPI cards, attrition by department, by satisfaction and overtime, and the leaver/stayer donut.

Step 4: a calculated column with DAX

We want to see joining trends by year, but the table has only `hire_date`. Add a **calculated column**: Table view (or Modeling) → **New column**:

DAX

```
Hire Year = YEAR ( hr_employees[hire_date] )
```

A calculated column is evaluated **row by row when the data loads** and stored in the model, just like a new column in Excel. Now a line/column chart of **headcount by hire year** works:

Python

```
hr["hire_year"] = hr.hire_date.dt.year
by_year = hr.groupby("hire_year").agg(hired=("emp_id", "count"), left=("left", "sum"))
by_year["attrition_%"] = (by_year.left / by_year.hired * 100).round(1)
by_year.tail(8)
```

OUTPUT

	hired	left	attrition_%
hire_year			
2019	82	16	19.5
2020	85	20	23.5
2021	120	22	18.3
2022	136	35	25.7
2023	196	60	30.6
2024	234	71	30.3
2025	284	93	32.7
2026	64	17	26.6

8 rows × 3 columns

Step 5: data groups and bins

Groups combine values into custom categories, such as the "age band" below, without changing the source data:

- **Bins** (numbers): right-click the field → **New group** → Group type *Bin* → bin size 5 → creates `age (bins)`.
- **List groups** (text): select several values → right-click → **Group** → rename, for example group *Mumbai* and *Pune* as "Maharashtra".

For flexible logic, a calculated column is more transparent:

DAX

```
Age Band =
SWITCH (
    TRUE (),
    hr_employees[age] < 25, "Under 25",
    hr_employees[age] < 35, "25-34",
    hr_employees[age] < 45, "35-44",
    "45 and over"
)
```

Python

```
bands = pd.cut(hr.age, [0, 24, 34, 44, 120], labels=["Under 25", "25-34", "35-44", "45 and over"])
hr.groupby(bands, observed=False).agg(employees=("emp_id", "count"), attrition_pct=("left", lambda s: round(s.mean() * 100, 1)))
```

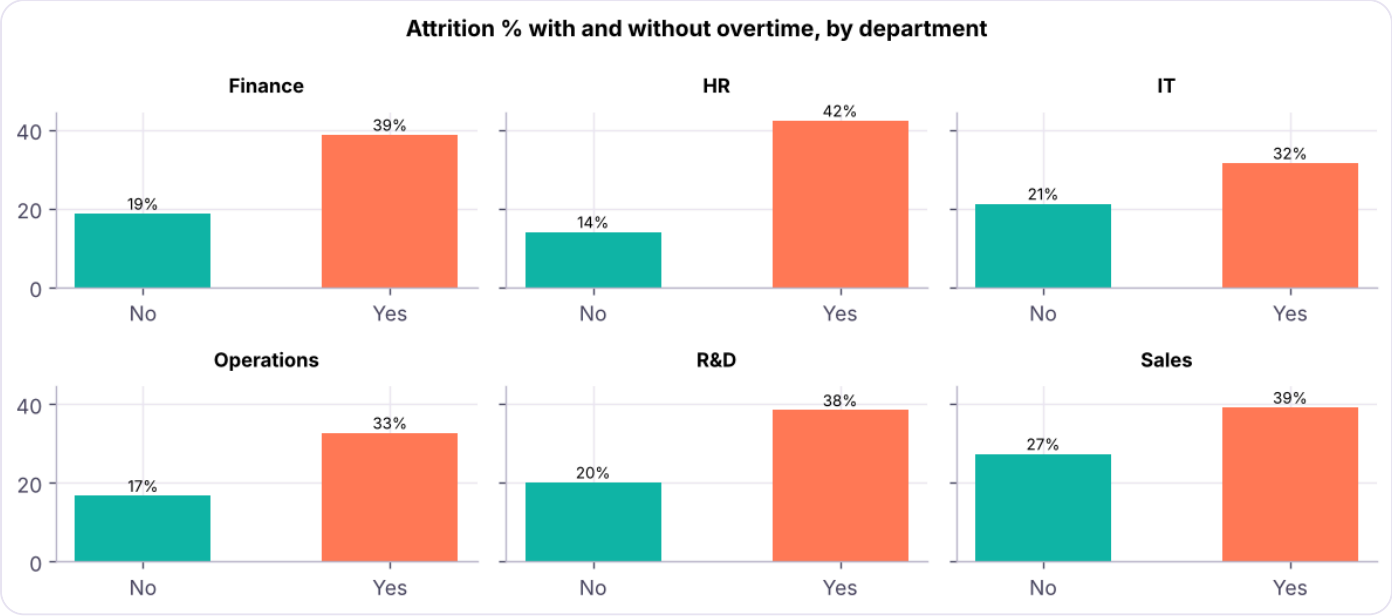
OUTPUT

	employees	attrition_pct
age		
Under 25	148	35.1
25–34	624	26.9
35–44	574	23.5
45 and over	154	22.1

4 rows × 2 columns

Step 6: small multiples

Small multiples split one visual into a grid of identical mini-charts by a field. Drag `department` into the visual's **Small multiples** well: you get one small column chart of attrition by satisfaction for every department. This lets you compare patterns across groups at a glance without a cluttered legend.



Small multiples: attrition % by overtime for each department. The same pattern repeats everywhere, which says overtime is a company-wide issue, not a one-department issue.

What does the report tell us?

```python
t = hr.groupby("overtime").left.agg(["count", "mean"]); t["mean"] = (t["mean"] *
100).round(1)
t.columns = ["employees", "attrition_%"]
print(t.to_string()); print()
low = hr[hr.job_satisfaction ⩽ 2].left.mean() * 100; high = hr[hr.job_satisfaction ⩾
3].left.mean() * 100
print(f"Attrition for satisfaction 1-2: {low:.1f}%   vs   3-4: {high:.1f}%")
```

OUTPUT

```
           employees  attrition_%
overtime
No              1067         21.6
Yes              433         36.7

Attrition for satisfaction 1-2: 32.5%   vs   3-4: 19.4%
```

Insight sentences for a manager: *overtime workers leave far more often than others; low satisfaction scores go with higher attrition; Sales has the highest attrition among departments*. Notice the careful wording: these are **associations** in the data, not proof of cause. (The ML track shows how to test them properly.)

Good habits from day one

1. **Name things clearly**: `Attrition %`, not `Measure 3`; rename pages ("HR Overview").
2. **Put measures in a dedicated table** later (Lesson 3) and format them (percentage, thousands separator).
3. **Check data types** in Power Query before loading.
4. **One idea per page**; the title should say the insight.
5. **Save often** (`.pbix`), and use clear version names.
6. **Do not use pie charts** with many slices; prefer bars (ranked) and use donuts only for 2–3 parts.

2026 UPDATE

Power BI in 2026

- **Copilot** can create a report page from a prompt ("Build an attrition overview for HR"), write DAX measures and summarise a page. It needs a paid Fabric/Premium capacity and your data to be well-modelled and named.
- **Power BI Project (PBIP)** format stores the report as text files, so teams can use **Git** and code review.
- **TMDL** (Tabular Model Definition Language) and **DAX query view** make models easier to script and test.
- **Direct Lake** (Fabric) reads Delta tables directly with import-like speed. (See Lesson 6.)

Summary

KEY TAKEAWAYS

- Power BI = Desktop (build) + Service (share) + Mobile (consume), now inside Microsoft Fabric.
- ETL: **Extract and Transform in Power Query, Load into the model**; analytics stages run from question to decision.
- Three views: **Report**, **Table**, **Model**; build with the Fields, Visualizations and Filters panes.
- A **card** shows a KPI; bar/column charts compare categories; a donut shows 2–3 parts; a matrix is a pivot table; slicers filter.
- A DAX **calculated column** (`YEAR`, `SWITCH`) is computed row by row at load time; **groups/bins** categorise without changing the data; **small multiples** repeat a chart per category.

Quick quiz

6 questions · tap an answer to check it

1. In Power BI, where do you perform the Extract and Transform steps of ETL?

A Report view

B Power Query

C The Service

D The Model view

Answer: B. Power Query connects to sources and transforms data; Close & Apply loads it into the model.

2. Which view shows relationships between tables?

A Report view

B Table view

C Model view

D Filters pane

Answer: C. Relationships are created and edited in Model view.

3. Power BI Desktop is available for:

A Windows and Mac

B Windows only

C Mac only

D Browsers only

Answer: B. Desktop is Windows-only; Mac users use the Service or a Windows virtual machine.

4. What does a calculated column do?

A Runs only when a slicer changes

B Computes a value for every row when the data is loaded and stores it in the model

C Replaces a measure

D Changes the source file

Answer: B. Calculated columns are row-by-row and stored; measures calculate on the fly.

5. Which visual repeats the same chart for every value of a field?

A Matrix

B Slicer

C Small multiples

D Card

Answer: C. Small multiples create a grid of mini-charts split by a category.

6. Which licence do you need to share a report with colleagues who also have Pro licences?

A None, Desktop is enough

B Free Service licence

C Power BI Pro (or Premium Per User)

D Gateway licence

Answer: C. Sharing in the Service requires Pro for creators and viewers (unless the workspace is on Premium/Fabric capacity).

EXERCISE

Build the HR page

Build the page shown above in Power BI Desktop with `hr_employees.csv`. Then add: (1) a slicer for `city` and `gender`; (2) a matrix of attrition % by `job_title` and `education`; (3) an `Age Band` column and a column chart of attrition % by band; (4) a title text box that states your main finding.

Show solution (worked answer)

(1) Slicer visual, fields `city`, `gender` (Format → Slicer settings → Style: Tile or Dropdown). (2) Matrix: Rows `job_title`, Columns `education`, Values `Attrition %` (turn on Cell elements → Background colour for a heat map). (3) Use the `SWITCH` formula in this lesson; chart X-axis `Age Band`, Y-axis `Attrition %`; sort the axis in the right order using a sort-by-column. (4) Insert → Text box, for example "Overtime is linked to about 15 more attrition points company-wide." Verify the claim against your visuals before you publish.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between Power BI Desktop and Service? Desktop is the free authoring tool for models and reports on Windows; the Service is the cloud platform to publish, share, refresh on a schedule, build dashboards and apps, and apply security.

Q: What is ETL and where is it done in Power BI? Extract, Transform, Load: Power Query connects to sources and applies transformation steps; the result is loaded into the VertiPaq in-memory model.

Q: Calculated column vs measure? A calculated column is evaluated per row at refresh time and stored (uses memory); a measure is evaluated at query time in the current filter context (uses CPU). Use measures for aggregations such as sums and ratios.

Q: Report vs dashboard? A report is multi-page and interactive, built on one model. A dashboard is a single page of pinned tiles from several reports, available only in the Service.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.